

ENSO-Aware Adversarial Forecasting: A Unified WGAN-GP Framework for Extreme-Aware Imputation and Forecasting of Heatwave Days in Regional Temperature Time Series

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Abstract—Weather and climate time series are simultaneously non-stationary, gap-prone, and characterized by rare but high-impact extreme events such as heatwaves. Over India, the frequency and persistence of heatwaves are further modulated by the El Niño–Southern Oscillation (ENSO), so that the days of greatest practical consequence cluster in El Niño-affected seasons. Existing generative adversarial network (GAN) approaches to this domain address imputation and forecasting as separate problems, ignore large-scale climate drivers such as ENSO, and existing extreme-value-aware adversarial objectives have been demonstrated on financial series or on continental-scale spatial climate fields rather than on region- or city-scale time series. We propose a unified adversarial framework in which a single ENSO-conditioned generator performs both missing-data imputation and multi-step forecasting of heatwave days for a regional temperature series, trained with a Wasserstein GAN with gradient penalty (WGAN-GP) objective augmented by a generalized-extreme-value-inspired exceedance-weighted loss term. The generator and critic are conditioned on a causally lagged Oceanic Niño Index, so that the model can learn how the likelihood and intensity of heatwave days shift with the ENSO phase. The framework is evaluated under a temporally held-out train/validation/test protocol using both continuous-value metrics (RMSE, MAE, R^2 , extreme-tail RMSE) and event-level detection metrics (probability of detection, false alarm ratio, critical success index) aligned to the India Meteorological Department heatwave definition, reported separately for El Niño, La Niña and neutral periods, enabling direct comparison with classification-based heatwave studies. **[TODO: Replace with a 2–3 sentence summary of the final quantitative results once real-data experiments are complete, e.g. “On N years of ERA5-Land data for Chennai, the proposed framework improves heatwave-event CSI by $X\%$ over the strongest classical baseline during El Niño periods while remaining competitive on bulk RMSE.”]**

Index Terms—generative adversarial networks, WGAN-GP, time series imputation, weather forecasting, extreme value theory, heatwave prediction, El Niño–Southern Oscillation, ERA5-Land

station coverage all produce gaps that must be filled before a series can be analyzed or modeled, while downstream applications such as early-warning systems require genuine forecasts of future values. Both problems are compounded when the quantity of interest is prone to extreme excursions: temperature time series in tropical, heatwave-prone regions such as peninsular India are dominated in aggregate by smooth seasonal and diurnal cycles, yet the events of greatest practical consequence—heatwaves—sit in the tail of the distribution, precisely where models trained under a mean-squared-error objective are known to be least reliable.

[TODO: Expand this paragraph with 1–2 sentences on the human/economic impact of heatwaves in India specifically (cite a recent IMD or health-impact source), to motivate the practical stakes before moving to the technical gap.]

Heatwave occurrence over India is not stationary from year to year. The El Niño–Southern Oscillation (ENSO), the dominant mode of interannual climate variability in the tropical Pacific [1], [2], influences Indian surface temperature through atmospheric teleconnections, and the frequency and duration of Indian heatwaves have been linked to warm (El Niño) conditions in the central and eastern Pacific [3], [4]. This relationship is itself non-stationary [5], which makes it a natural candidate for data-driven rather than fixed parametric modeling. A model that imputes or forecasts temperature without access to the ENSO state must treat El Niño and La Niña years as draws from a single distribution, and is therefore likely to under-predict exactly the heatwave days that cluster in El Niño-affected seasons.

Generative adversarial networks (GANs) [6] have been applied productively to both imputation [7], [8] and forecasting/generation [9]–[11] of weather and climate data, and a separate line of work has shown that explicitly reshaping the training objective around extreme value theory (EVT) improves tail fidelity in both non-adversarial [12], [13] and adversarial [14]–[17] settings. As detailed in Section II, however,

I. INTRODUCTION

Missing observations and the need to project future states are two persistent, related challenges in regional climate time series analysis. Sensor outages, transmission loss, and irregular

no existing framework combines imputation and forecasting within a single adversarially-trained generator for a region-scale series, no extreme-value-aware GAN has been applied to city- or region-scale temperature imputation/forecasting in a heatwave-prone setting such as India, and none conditions on the ENSO state that modulates heatwave risk. A recent, closely related study on Chennai heatwave classification [18] explicitly names time series GANs as a plausible, unexplored strategy for handling the class imbalance inherent to rare heatwave events, but does not pursue it.

This paper makes the following contributions:

- 1) A unified adversarial framework in which a single generator performs both interior-gap imputation and multi-step forecasting, differing only in where the reconstruction mask falls within a training window.
- 2) ENSO conditioning of both the generator and the WGAN-GP critic on a causally lagged Oceanic Niño Index (ONI), so that imputed and forecast heatwave days are generated conditionally on the El Niño, La Niña or neutral state of the tropical Pacific.
- 3) A generalized-extreme-value-inspired exceedance-weighted loss term that makes the reconstruction objective explicitly sensitive to tail (heatwave) behavior, combined with a WGAN-GP adversarial objective [19], [20].
- 4) A rigorous, temporally held-out evaluation protocol—avoiding a data-leakage failure mode we identified in early iterations of this work (Section IV)—reporting both continuous and event-level (POD/FAR/CSI) metrics, stratified by ENSO phase, for direct comparability with classification-style heatwave studies.
- 5) **[TODO: An empirical evaluation on __ years of ERA5-Land data for __ (city/region), showing __.]**

II. RELATED WORK

A. Statistical and Deep Learning Approaches to Weather Time Series

Classical statistical models such as ARIMA and SARIMA remain common baselines for climate time series owing to their interpretability and modest data requirements, but they assume linear, stationary dynamics and struggle with the non-linear, non-stationary behavior typical of real weather data [21]. Recurrent architectures such as LSTM and hybrid CNN-LSTM networks capture non-linear temporal dependencies directly from data, but models trained under a mean-squared-error objective regress toward the conditional mean, systematically under-representing tail events such as heatwaves [21]. This smoothness bias motivates the shift toward generative and distribution-aware modeling reviewed below.

B. Generative Adversarial Approaches to Climate and Weather Data

GANs have been applied to weather and climate data along two largely separate tracks: imputation, and generation/forecasting. On the imputation side, Luo et al. [7] introduced an early GAN-based framework for multivariate time

series imputation, and Zhao et al. [8] added a self-attention discriminator for imputing aviation time series; neither targets weather-specific tail behavior, and both evaluate imputation in isolation from any downstream forecasting task. On the generation/forecasting side, Bihlo [9] trained a conditional deep convolutional GAN on ERA5 reanalysis to forecast 24-hour-ahead geopotential height, 2 m temperature, and precipitation over Europe; Balogun et al. [10] instead perform unconditional stochastic generation of hourly 2 m temperature fields conditioned on month, region, and time-of-day; and Glawion et al. [11] downscale ERA5 precipitation to kilometer and sub-hourly resolution with explicit evaluation of extreme rainfall preservation. Across this body of work, imputation and forecasting are treated as separate problems solved by separately trained generators, conditioning is limited to calendar or regional information rather than large-scale climate drivers such as ENSO, and none are evaluated on India or on heatwave-prone tropical climates.

C. Extreme-Value-Aware and Tail-Focused Generative Modeling

A parallel line of research reshapes the training objective around the tail of the distribution rather than its mean. Ding et al. [12] proposed an extreme value loss for time series prediction that reweights the objective using a fitted extreme value distribution, later generalized by Zhang et al. [13]. Within GAN architectures, the Pareto-GAN of Huster et al. [14] extends generator representational capacity into heavy-tailed regimes using extreme value theory, and the ExGAN of Bhatia et al. [15] conditions sample generation directly on user-specified extremeness. Boulaguiem et al. [16] combine EVT with a GAN discriminator for spatial extremes, and the most climate-specific recent example, the DeepX-GAN of Liu et al. [17], embeds a spatial extremal-dependence metric into the discriminator to generate physically plausible unseen heat extremes across the Middle East and North Africa. This family of work establishes that tail-aware adversarial objectives measurably outperform vanilla adversarial or MSE-only training on extreme-event fidelity—but every example operates on financial data, generates unconditional spatial fields, or targets continental-scale spatial extremes, rather than gap-filling and forecasting a specific city-scale time series.

D. Heatwave-Specific Prediction, with Emphasis on India

Jacques-Dumas et al. [22] trained a CNN on climate-model output with class-undersampling and transfer learning to forecast long-lasting heatwaves, addressing the severe class imbalance inherent to rare-event prediction. Alhusainan et al. [23] combined ConvLSTM, transformer-based temporal attention, and physics-informed constraints to forecast extreme heat in Kuwait from hourly ERA5-Land data, using ARIMA/SARIMA as classical baselines—a design closely related in spirit, though not architecture, to the present work. Within India specifically, Ratnam et al. [24] used machine learning to predict maximum temperatures ten days ahead,

and—most directly relevant—Choudary et al. [18] benchmarked nine machine learning classifiers for next-day heatwave event classification on fourteen years of Chennai (Tambaram) station data, reporting a best accuracy of 98.92% with a CNN. Qureshi et al. [25] developed a seasonal-adjusted hybrid time series model for heatwave warning under a similar rare-event framing. Notably, Choudary et al. [18] explicitly identify time series GANs as a plausible strategy for generating temporally coherent synthetic minority-class sequences to address class imbalance, but do not implement or evaluate the approach, leaving it as an open direction that this work pursues.

E. ENSO and Indian Heatwaves

ENSO is conventionally monitored through sea surface temperature (SST) anomalies in the Niño-3.4 region [1]; the NOAA Oceanic Niño Index (ONI), a three-month running mean of ERSSTv5 Niño-3.4 anomalies [26], [27], defines El Niño (La Niña) episodes as periods in which the ONI is at or above $+0.5^{\circ}\text{C}$ (at or below -0.5°C) for at least five consecutive overlapping seasons. Rohini et al. [3] showed that heatwave frequency and duration over India have increased and are associated with positive SST anomalies in the central and eastern Pacific, and Ratnam et al. [4] related Indian heatwaves to large-scale circulation anomalies forced from the tropical oceans. The ENSO–Indian climate link has also weakened and strengthened over recent decades [5]. Despite this physical motivation, the data-driven heatwave studies reviewed above do not use the ENSO state as a model input, and none evaluate skill separately in El Niño and La Niña years.

F. Research Gap

Four gaps emerge from this survey. First, GAN-based imputation and GAN-based forecasting/generation have developed as separate literatures; no framework performs both within a single adversarially-trained generator for the same regional dataset. Second, extreme-value-aware adversarial objectives have been demonstrated on financial series or on continental/global spatial climate fields, but not on a city- or region-scale time series task that combines imputation and forecasting. Third, the most directly comparable India-focused study explicitly flags time series GAN generation as an unexplored strategy for heatwave-related class imbalance, without pursuing it [18]. Fourth, although ENSO is a documented driver of Indian heatwave variability [3], [4], no generative imputation or forecasting model for Indian temperature conditions on the ENSO state or reports its skill by ENSO phase. This work addresses all four.

III. DATA

[TODO: Fill in final data description once the full pull is complete. Draft below based on the Review-1 pipeline, extended per the revised scope.]

A. Regional Temperature Series

Daily meteorological data were obtained from the ERA5-Land reanalysis product [28], [29] (ECMWF/ERA5_LAND/DAILY_AGGR) via the Google Earth Engine API [30], for a point located at **[TODO: 13.08°N, 80.23°E (Chennai)] / [TODO: list any additional cities if the multi-site extension is pursued]**, covering **[TODO: 2010-01-01 to 2024-12-31 (15 years). Note: the ENSO-stratified evaluation needs several El Niño and La Niña episodes in each of the training and test periods; the 1950–2025 extract already in the project repository is better suited to this than a 15-year window.]** Variables extracted include 2 m air temperature (temperature_2m), surface pressure, total precipitation, and **[TODO: any additional variables: dewpoint temperature, wind components, etc.]**. **[TODO: Because the IMD heatwave definition is stated in terms of daily maximum temperature, also extract temperature_2m_max and use it as the heatwave target.]** Temperature values were converted from Kelvin to Celsius. The series was split chronologically into training (**[TODO: 2010–2020]**), validation (**[TODO: 2021–2022]**), and test (**[TODO: 2023–2024]**) partitions, with all scaling parameters and extreme-value thresholds fit on the training partition only to avoid information leakage from validation/test periods into model development.

B. ENSO Index

The ENSO state is represented by the monthly ONI [27], computed from ERSSTv5 Niño-3.4 SST anomalies [26]. Each calendar month is labeled El Niño, La Niña, or neutral using the NOAA episode rule given in Section II, and the monthly ONI is mapped to daily resolution by assigning each day the value of its month. Because the ONI is a centered three-month mean, the value for a given month uses SST from the following month; to keep the forecasting task operationally causal, the conditioning variable for a window starting on day t_0 is the most recent ONI value that would have been published before t_0 , i.e., the ONI lagged by $\ell \geq 2$ months. **[TODO: Report the number of El Niño, La Niña and neutral months in the training, validation and test partitions.]**

C. Heatwave Days

Heatwave days are labeled following the India Meteorological Department (IMD) criteria [31]. For a coastal station such as Chennai, a heatwave may be declared when the daily maximum temperature departs from its calendar-day climatological normal by at least 4.5°C and the actual maximum temperature is at least 37°C ; a departure of more than 6.4°C denotes a severe heatwave. Calendar-day normals are computed from the training partition only. **[TODO: ERA5-Land is a gridded reanalysis and may be biased relative to station maxima; state whether the absolute 37°C condition is kept, bias-corrected against IMD station data, or replaced by a training-period percentile threshold, and report the resulting number of heatwave days per partition and per ENSO phase.]**

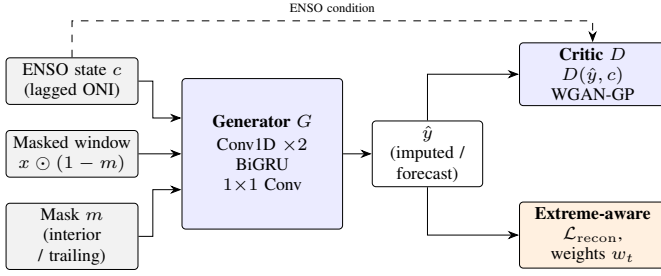


Fig. 1. Proposed ENSO-aware framework. A single generator receives a masked temperature window, the mask, and the lagged ENSO state, and fills either an interior gap (imputation) or the trailing k days (forecasting). It is trained with an ENSO-conditioned WGAN-GP critic and an extreme-aware weighted reconstruction loss.

IV. METHODS

A. Problem Formulation

We formulate imputation and forecasting as a single masked-reconstruction problem. Given a window of length W drawn from the scaled temperature series, a binary mask $m \in \{0, 1\}^W$ indicates which positions are hidden from the model. When the masked span sits in the interior of the window, the task is imputation; when the masked span is the trailing k steps of the window with no observations beyond it, the task is equivalent to a k -step-ahead forecast. During training, the mask type (interior vs. trailing), position, and length are resampled independently for every window, which is the key methodological safeguard against the failure mode we identified in early development of this project: in an earlier iteration, a single fixed masked block was used for both training and evaluation, so the generator was directly supervised, over hundreds of training epochs, to reconstruct the exact segment it was later scored on, while classical baselines (ARIMA, SARIMA, KNN) never had access to it. This produced implausible results (generator $R^2 \approx 0.99$ against negative- R^2 classical baselines) that we do not consider valid and have discarded in favor of the protocol described here.

Each window is accompanied by a conditioning vector $c \in \mathbb{R}^{W \times d_c}$ that carries the lagged ONI described in Section III and, optionally, a sine-cosine encoding of the day of the year. The generator therefore learns the conditional distribution $p(y_{\mathcal{M}} \mid y_{\bar{\mathcal{M}}}, c)$ of the hidden values $y_{\mathcal{M}}$ given the observed context $y_{\bar{\mathcal{M}}}$ and the ENSO state, rather than a single distribution pooled over all ENSO phases.

B. Generator and Critic Architecture

The generator is a shared backbone consisting of two 1-D convolutional layers (kernel size 5) over the input channels (masked value, mask indicator, and the ENSO conditioning channels c), followed by a bidirectional GRU and a final 1×1 convolution projecting back to a single output channel. The same generator, without architectural modification, is used for both imputation and forecasting windows.

The critic follows the Wasserstein GAN with gradient penalty (WGAN-GP) formulation [19], [20]: a convolutional

critic outputs an unbounded realism score rather than a bounded probability, and a gradient penalty term enforces the Lipschitz constraint required for the Wasserstein distance approximation to be meaningful. Following the conditional GAN formulation [32], the critic receives the conditioning channels c concatenated with the completed window, so that it judges whether a heatwave sequence is realistic *for the given ENSO state*. The critic loss is

$$\mathcal{L}_D = \mathbb{E}[D(\hat{x}, c)] - \mathbb{E}[D(x, c)] + \gamma \mathbb{E}[\|\nabla_{\tilde{x}} D(\tilde{x}, c)\|_2 - 1]^2, \quad (1)$$

where x is a real window, $\hat{x}$ the generator-completed window, $\tilde{x}$ a random interpolation between the two, and γ the gradient-penalty weight. This was a deliberate design choice over the vanilla discriminator used in early iterations of this work, where the adversarial loss term was weighted more than three orders of magnitude below the reconstruction loss term and therefore contributed negligibly to training—in that configuration the model was, in substance, a supervised reconstruction network rather than a GAN.

C. Extreme-Value-Aware Loss

To make the reconstruction objective sensitive to tail behavior, we introduce an exceedance-weighted mean-squared-error term inspired by the extreme value loss of Ding et al. [12] and its generalization by Zhang et al. [13]. Let τ_t denote an extreme-value threshold for day t (e.g., the 95th percentile of the training distribution, or the IMD heatwave threshold of Section III, i.e., the calendar-day normal plus 4.5°C), and σ the training standard deviation. For a true value y_t and generator output $\hat{y}_t$ at masked position t , we define a per-timestep weight

$$w_t = 1 + \lambda \cdot \max\left(0, \frac{y_t - \tau_t}{\sigma}\right), \quad (2)$$

and the reconstruction loss is the w_t -weighted mean squared error over masked positions,

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{M}|} \sum_{t \in \mathcal{M}} w_t (\hat{y}_t - y_t)^2, \quad (3)$$

where $\mathcal{M}$ is the set of masked positions in a batch and λ controls the strength of the extreme-event emphasis. This is a deliberately simplified, differentiable approximation of the full GEV-likelihood-based reweighting in [12] and [13]; we describe it here as an adaptation rather than a reproduction, and note that fitting a full generalized Pareto or GEV distribution to the residual exceedances is a natural extension for future work (Section VI).

The full generator objective combines the adversarial term with this weighted reconstruction term:

$$\mathcal{L}_G = -\mathbb{E}[D(\hat{x}, c)] + \beta \mathcal{L}_{\text{recon}}, \quad (4)$$

with β chosen so that the adversarial and reconstruction terms are of comparable magnitude ([**TODO: report final β , λ , γ values used**]).

D. ENSO-Balanced Training

El Niño months are a minority of the record, and heatwave days are a minority within them. To keep the generator from fitting mainly the neutral-phase climatology, training windows are drawn with probabilities that equalize the three ENSO phases within each mini-batch, while validation and test windows are left at their natural frequencies. **[TODO: Confirm whether phase-balanced sampling is used in the final runs, and report its effect in the ablation table.]**

E. Baselines and Ablations

ARIMA, SARIMA, and k -nearest-neighbor (KNN) baselines are fit per window using only the observed (non-masked) context—for interior gaps, the pre- and (where applicable) post-gap context; for trailing (forecast) gaps, only the pre-gap context, exactly matching the information a deployed forecasting system would have. All baselines are scored on the identical held-out windows as the proposed model. We additionally report the following ablations, each retaining the generator architecture:

- *No adversarial term* ($\beta \rightarrow \infty$ equivalent, i.e. reconstruction loss only, no critic), to isolate the marginal contribution of the adversarial objective itself;
- *No ENSO conditioning*, in which c is removed from both the generator and the critic;
- *Shuffled ENSO*, in which the ONI series is permuted across years, as a control showing whether any gain comes from the ENSO signal itself rather than from the extra input channel;
- *ONI lag*, varying $\ell \in \{2, 3, 6\}$ months, to measure how far ahead the ENSO state carries useful information about heatwave days.

[TODO: Add LSTM baseline results if included in the final comparison.]

F. Evaluation Protocol

We report continuous-value metrics—RMSE, MAE, R^2 , and an extreme-subset RMSE restricted to true values exceeding τ_t —alongside event-level detection metrics standard in operational meteorological verification: probability of detection ($\text{POD} = TP/(TP + FN)$), false alarm ratio ($\text{FAR} = FP/(TP + FP)$), and critical success index ($\text{CSI} = TP/(TP + FP + FN)$), where an “event” is a masked day that meets the heatwave criterion of Section III. Where possible, τ_t is aligned with the India Meteorological Department’s operational heatwave definition (departure from the calendar-day climatological normal, rather than a fixed percentile) once sufficient years of data are available to compute reliable normals. All metrics are reported over the full test period and separately for El Niño, La Niña, and neutral months, since a model may improve heatwave detection in El Niño periods while leaving its aggregate score almost unchanged. Differences in forecast error between the proposed model and each baseline are tested with the Diebold–Mariano test [33]. This choice of metrics allows direct comparison with classification-based heatwave studies such as [18] and

[25], which report precision/recall-style scores rather than continuous error.

V. RESULTS

[TODO: This entire section is a template pending real-data experiments. Replace Tables I–III with the actual test-set numbers, run across multiple random seeds, and report mean $\pm$ std. Do not submit placeholder numbers—the tables only show the expected format and are explicitly not experimental results.]

[TODO: Add: (i) a qualitative figure showing a reconstructed/forecasted heatwave episode in an El Niño year against ground truth for the proposed model vs. the strongest baseline; (ii) a calibration or reliability plot for the extreme-value-weighted loss; (iii) an ablation figure isolating the effect of λ (extreme loss strength) on POD/CSI vs. bulk RMSE, since this trade-off is the paper’s central empirical claim; (iv) a figure of heatwave-day counts per year against the annual ONI, for observed, imputed and forecast series.]

VI. DISCUSSION

[TODO: Once real results are available, discuss: (1) whether the proposed model trades bulk accuracy for extreme-event detection, and whether that trade-off is favorable for a heatwave early-warning use case; (2) whether ENSO conditioning improves heatwave detection specifically in El Niño periods, whether the shuffled-ENSO control removes that gain, and which ONI lag is most informative; (3) how the results compare, qualitatively, to the classification-based framing in [18]; (4) limitations—single-city/single-variable scope, reliance on ERA5-Land reanalysis rather than in-situ station data, the small number of El Niño episodes in any test period, the non-stationary ENSO–monsoon relationship [5], and the simplified (non-GEV-likelihood) extreme loss; (5) concrete extensions: other climate drivers such as the Indian Ocean Dipole, multi-city/spatial modeling (cf. [34]), a full GEV-likelihood loss, and a diffusion-model comparison given the recent shift toward diffusion in generative weather modeling.]

VII. CONCLUSION

[TODO: 2–4 sentence summary once results are final: restate the gap, the proposed ENSO-aware framework, and the headline empirical finding.]

DATA AVAILABILITY

[TODO: State that ERA5-Land data are publicly available via the Copernicus Climate Data Store / Google Earth Engine, that the ONI is publicly available from the NOAA Climate Prediction Center, and specify how your processed dataset and train/val/test splits will be shared (e.g. a public repository link).]

TABLE I
TEST-SET RESULTS OVER THE FULL TEST PERIOD (MEAN $\pm$ STD OVER ≥ 3 SEEDS). POPULATE WITH REAL ERA5-LAND TEST-SET NUMBERS.

Model	RMSE	MAE	R ²	Extreme RMSE	POD	FAR	CSI
Proposed (ENSO-cond. WGAN-GP + EVT loss)	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Ablation (no adversarial term)	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Ablation (no ENSO conditioning)	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
ARIMA	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
SARIMA	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
KNN	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

TABLE II
HEATWAVE-DAY DETECTION BY ENSO PHASE ON THE TEST PERIOD (CSI, MEAN $\pm$ STD OVER ≥ 3 SEEDS).

Model	El Niño	Neutral	La Niña
Proposed	[TODO:]	[TODO:]	[TODO:]
No ENSO conditioning	[TODO:]	[TODO:]	[TODO:]
Shuffled ENSO	[TODO:]	[TODO:]	[TODO:]
Strongest classical baseline	[TODO:]	[TODO:]	[TODO:]
Heatwave days in test period	[TODO:]	[TODO:]	[TODO:]

TABLE III
ABLATION OVER THE ONI LAG ℓ AND THE EXTREME-LOSS STRENGTH λ (TEST PERIOD).

Setting	RMSE	Extreme RMSE	POD	CSI
$\ell = 2$ months	[TODO:]	[TODO:]	[TODO:]	[TODO:]
$\ell = 3$ months	[TODO:]	[TODO:]	[TODO:]	[TODO:]
$\ell = 6$ months	[TODO:]	[TODO:]	[TODO:]	[TODO:]
$\lambda = 0$ (plain MSE)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
$\lambda =$ [TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

CODE AVAILABILITY

[TODO: Link to a public repository (e.g. GitHub) containing the training and evaluation code.]

AUTHOR CONTRIBUTIONS

[TODO: Specify each author's contribution (conceptualization, data curation, methodology, writing, supervision, etc.), if required by the target venue.]

COMPETING INTERESTS

The authors declare no competing interests.

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